

thereby blocking vision. Laser beams easily pass through transparent portions of the eye, including the cornea and lenses, to reach the region where energy of the laser is absorbed and used for treatment.

Traditionally, the use of xenon arc lamps and sunlight to focus over the choroid coat was really painful and required high-level anesthesia. Introduction of high-pulsed lasers like Nd:YAG was a major breakthrough when dealing with eyes. When focused on a tiny spot at the detached retina in order to weld it to the choroid coat, it leads to painless surgery.

Lasers are also used to burn small tumors on the surface of the eye or its vessels. These are being used to treat coma, cataract, sealing of retina and viral diseases of the eye.

As lasers are less damaging than x-rays, these are extensively and

Professional bodies in laser and allied fields

1. Indian Laser Association
2. Photonics Society of India
3. Optical Society of India
4. IEEE-Photonics Society
5. IEEE-Laser and Electro-Optics Society
6. SPIE - The International Society for Optics and Photonics

effectively used to treat different types of cancers by removing those enlarged tissues that are responsible for the cancer.

Lasers are also effective in curing diseases of gynecology, ear, nose, throat, tongue, palate and cheeks. These are effective in reducing tumors, too.

Photodynamic therapy (PDT) is the new form of cancer treatment, which combines laser with light-sensitive dye, hemato-porphyrin derivative (HPD), which is derived

from cow's blood, and travels through the body of a patient and settles in malignant tissues. An argon red light is used to activate the HPD, which energises to release a highly-reactive chemical that destroys cancer cells. Reports say that PDT is 80 to 90 per cent successful in causing total regression of tumors.

Lasers are also used to treat colonic and gastro-intestinal cancers. Tools like endoscopes use laser energy to destroy neuro-plastic tissue while preserving the bowel wall integrity. Lasers also find a role in clearing kidney stones.

Lasers are used for dental treatment, too. Using a process called laser glazing to perform tooth decay clearance, a high-intensity laser is focused on the decayed tissues of the tooth. This destroys the infections in the affected areas in a fraction of a second. **EFY**



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